

1. Number of Records: 1,500 Customer records; 12,000 Transaction records.

2. Number of Columns: 12 Customer columns; 11 original Transaction columns + 4 engineered lookup/audit flag columns.

3. Missing Values: 0 explicit blanks; documented 96 "Unspecified" devices and 96 "Unkown" locations.

4. Duplicate Records: 0 duplicate rows; verified primary keys Customer_ID and Transaction_ID.

5. Incorrect / Inconsistent Values: Highlighted 96 "Unkown" spelling typos and Channel vs. Device routing conflicts.

6. Data-Type Issues: Text-to-date timestamp conversions and alphanumeric ID formatting.

7. Unusual Values: 866 Dormant and 224 Restricted account transactions identified.

8. Potential Outliers: Most transaction amounts in the bank are small to medium-sized everyday payments. However, there is a small group of very high transaction amounts. These large amounts belong mainly to Business (SME) and Premium customers who naturally transfer bigger sums of money. Because these are real, normal business transactions rather than mistakes, we kept them in the data and simply added a flag to identify them easily.

9. Dataset Relationship: The Customer and Transaction datasets connect to each other using the **Customer ID**. Every transaction in the transaction list belongs to a valid customer in the customer list, meaning the link between both files is 100% complete and accurate. On average, each customer performed around 8 transactions during the three-month period.

10. Any Issues That Could Affect Analysis

- **Inactive Account Activity:** Transactions occurred on *Dormant* and *Restricted* accounts, pointing to potential compliance or system security risks.
- **Channel vs. Device Mismatches:** Channel and device records show logical conflicts (e.g., Mobile App on an ATM). Channel performance should be analyzed using only the **Channel** column.
- **Location Typos:** Misspellings like "Unkown" split city groupings and need standardization to prevent broken charts.